

Data Representativeness: What Are We Missing?

Using Data Science to Rethink Societal Gaps



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Why Representativeness Matters



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Data drives decisions, but not all data reflects reality.

- Unrepresentative data leads to flawed models and harmful outcomes.
- Representativeness is key to fairness, accuracy, and trust.

Why Representativeness Matters in a Data-Driven World

- **Data-Driven Decisions:** Organizations increasingly rely on data to guide strategies, affecting workers, users, customers, etc.
- **Consequences of Bias:** Non-representative data can lead to systematic biases, flawed models, and lost opportunities.
- **Real-World Impacts:** From product to design to public health, biased data has measurable consequences on people's lives.



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What is Data Representativeness?

- A dataset is **representative** when it **reflects** the **population** or **phenomenon** it models. It **mirrors** the characteristics of the target population.
- **Key factors:** Coverage, Diversity, Balance.
- **Challenges:** Bias, missing data, data collection limitations that may distort representativeness.
- **Sampling Methods:** Random and stratified sampling may be used to ensure statistical diversity and precision.



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Representativeness in AI systems

It **influences** fairness, accuracy, and reliability of models.

It **involves** creating datasets and models that accurately reflect the diversity of the target population, which is **essential** for drawing **valid** inferences and making **equitable** decisions.

It **ensures** that the models trained can **generalize** well to unseen data and make **fair** and **unbiased** decisions.

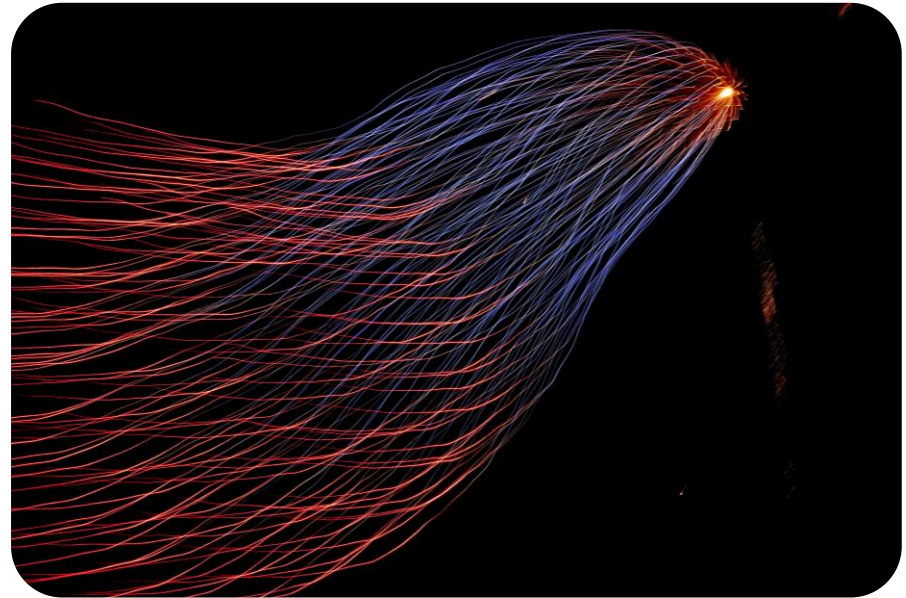


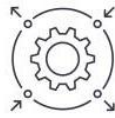
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Importance of Representativeness in AI



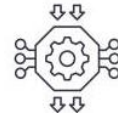
Fairness

Crucial for fairness in AI systems



Inclusivity

Ensures that the models do not favor any group or do not disproportionately disadvantage underrepresented groups



Generalizability

Less biases in training data leads to less biased and more accurate outcomes, better generalization

Implications for AI systems Development

Lack of Representativeness:

- **unfair** and discriminatory outcomes, especially for minority groups
- leads models to make judgments based on **stereotypes** rather than statistical evidence
- **perpetuated** by historical discrimination, selection biases, and the complexity of demographic variables

Representativeness:

- **influences** the design of algorithms
- **influences** the evaluation of algorithms performance
- **mitigates** biases
- **promotes** fairness
- implies better **generalizability**



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Challenges of Representativeness in AI



Bias and Limitations

Inherent bias in input data
and availability of diverse
datasets



Complexity of Representation

Ensuring that a dataset covers
the input space while also
mimicking the distribution of
the target population



Ethical and Privacy

Balancing data collection with
privacy concerns

Challenges for AI systems Development

Imbalanced Data Distribution: some real-world problems are naturally imbalanced with a long-tailed distribution, where common classes dominate and rare classes are underrepresented

Bias and Fairness: inherent biases in input data can lead to biased models, it is crucial to ensure the correct balancing of the coverage of the input space mimicking the distribution of the target population

Missing Data: it can significantly hinder the learning of accurate representations

Scalability and Efficiency: in large-scale systems efficient representation learning is necessary to handle accurate predictions ensuring a proper answer to different queries

Fairness in Representation: ensuring fairness in data representation is critical, especially when sensitive attributes are involved



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Approaches to Improve Representativeness in AI



Frameworks

to guide the creation of
representative datasets and
evaluations



Documentation

describing the dataset and
problem characteristics,
essential transparency and
accountability



Sparsity and Diversity

to ensure that the selected
samples are diverse and
mimics the population

How to Improve Representativeness in AI systems?



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Representative AI Evaluations

Develop tailored frameworks to ensure each part of the AI lifecycle (data collection, evaluation, deployment) promotes representativeness.

Highlight the trade-offs: balancing privacy concerns with the need for inclusive datasets

Dataset Documentation and Design

Use structured frameworks and questions to assess representativity in datasets

Differentiate between samples that cover the input space versus those that match the distribution of the target population

Adaptive Recruitment in Data Collection

Dynamically allocate resources to underrepresented groups or regions during data gathering, improving cohort balance across race, gender, age, and ethnicity

Representation Engineering

Focus on high-level population patterns instead of individual patterns

Enables better control and understanding of how models represent different groups

Audit Tools for Stereotype Dimensions

Measure how much specific traits dominate group stereotypes in models

Address bias not just in sentiment but in content emphasis

From Data to Decisions:

Does Bias in AI Reflect Bias in Society?



The Cost of Skewed Data – When Bias Becomes a Liability



Model Degradation

Model fail to generalize, especially when applied outside its biased sample



Systemic Risks

Critical systems risk to produce unfair outcomes, since biased predictions disproportionately affect marginalized groups



Societal Consequences

Data bias can amplify inequalities in healthcare, finance, education, justice, conducting to ethical and social consequences

When Data Gaps Hurt People

Facial Recognition Bias:

Misidentifies women and minorities.

Medical Trials Bias:

Female underrepresentation leads to health disparities.

Crash Test Bias:

Cars designed with male physiology in mind; women are 47% more likely to be seriously injured in accidents.



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Algorithms and Society: Are We Reinforcing the Same Biases?

Conceptual Understanding

Representativeness in AI is not just about statistical accuracy but also **involves inclusiveness and fairness**. It is a foundational concept that has been debated in various fields, and is now central to discussions on fairness and accountability in AI.

Data Representation

Ensuring **diverse** representation in AI datasets is critical for the models to function **effectively** across different demographic groups. This includes considerations of race, gender, age, and socioeconomic status.

Stereotypes and Bias

In language models, representativeness affects how **stereotypes are formed and perpetuated**. The directionality and content of stereotypes can differ, impacting the perceived fairness of AI systems. This highlights the need for careful auditing and debiasing to address both the direction and content of stereotypes.

Social Implications

The concept of representativeness in AI also ties into broader societal issues. AI systems can reflect societal biases **if not carefully designed and evaluated**, potentially **reinforcing existing inequalities**.

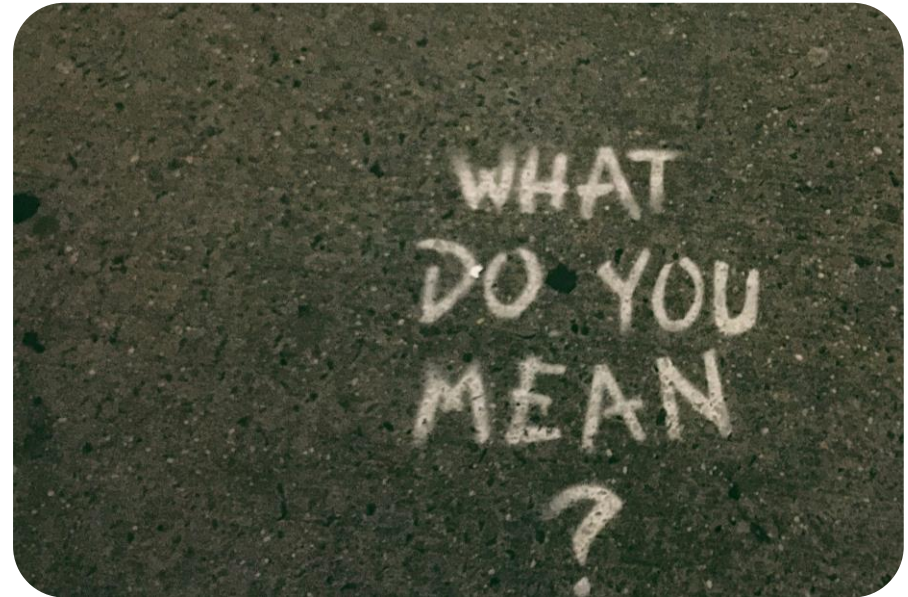


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From Data to Society



Biased Datasets vs Biased Systems

Systemic exclusion in datasets mirrors structural inequalities in society



The Gender Gap as a Data Problem

Underrepresentation of women in datasets impacts AI outcomes



Our Challenge

How might your models reflect or even reinforce societal gaps?
What are you missing?

Biased training data → *biased models*.

Biased leadership structures → *biased outcomes*.

We wouldn't accept biased datasets in AI →
why accept them in society?



Rethinking Representation: Data and Beyond

Data Inclusivity

Audit datasets regularly for representation gaps. Adjust sampling strategies and feature engineering accordingly.

Cross-Disciplinary Awareness

Collaborate with ethicists, sociologists, minorities and marginalized communities to contextualize data. Question whose data, voices, and perspectives are missing.

Representation Beyond Data

Advocate for representation in teams, leadership and policies that shape data practices.



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What Are We Still Missing?

Incomplete Mirrors

- Our datasets, like our systems, often reflect only part of the whole.
- What is left unseen?
- Document, quantify, describe, propose solutions.

Beyond the Numbers

- Bias is not always in the model.
- It starts with the problem framing and who is excluded from it.
- Question, advise, propose.

Ongoing Vigilance

- Representation is not a one-time fix.
- It is a continuous, ethical commitment.
- Be aware, discuss, be interventive.

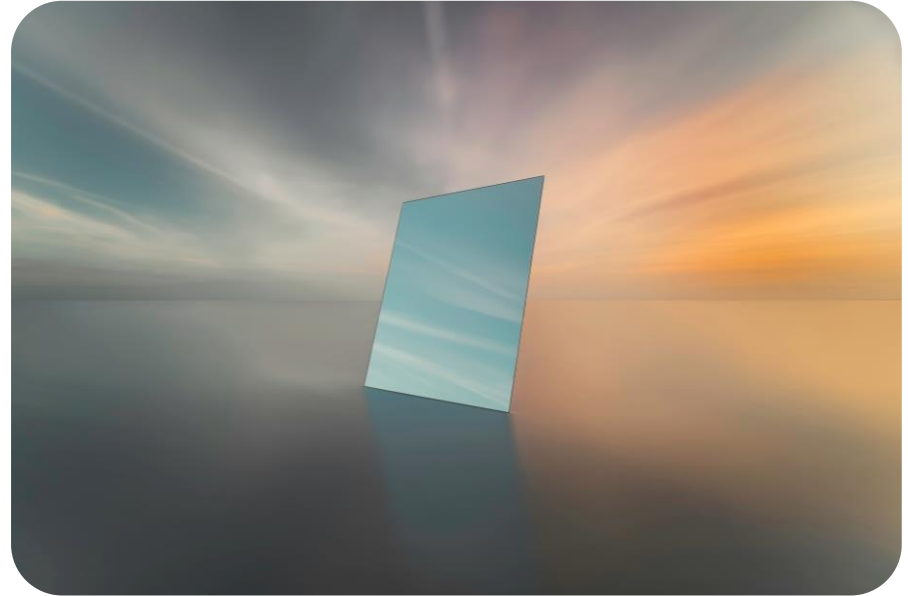


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**Representation, inclusion,
and diversity are not just
technical requirements.**

They are societal imperatives.

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